David Luca Villringer

Curriculum vitae

Institute of Mathematics, EPFL
Lausanne, Switzerland
Citizenship: German

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Research interests

Deterministic and stochastic partial differential equations in fluid mechanics: mixing, enhanced dissipation, hydrodynamic and magnetohydrodynamic instabilities, and dynamo theory.

Academic appointment

Sep 2026–present **Postdoctoral Researcher** — EPFL
Chair of Mathematical Analysis, Calculus of Variations and PDEs.

Education

2022–2026 **PhD in Pure Mathematics** — Imperial College London
Supervisors: Michele Coti Zelati and Martin Hairer.
CDT in Mathematics of Random Systems.

2021–2022 **MASt in Pure Mathematics** *Distinction* — University of Cambridge
Thesis: *Singular Limits of Hyperbolic PDEs*. Supervisor: Claude Warnick.

2018–2021 **BSc in Mathematics** *First Class Honours* — University College London

Published and accepted papers

2026 V. Navarro-Fernández and D. Villringer. *Spectral instability in the smooth Ponomarenko dynamo*. *Annals of PDE* **12** (2026), 32. doi:10.1007/s40818-026-00255-1.

2026 V. Navarro-Fernández and D. Villringer. *Nonlinear instability for the 3D MHD equations around the Taylor–Couette flow*. *Nonlinearity* **39** (2026), 055009. doi:10.1088/1361-6544/ae65fc.

Accepted M. Coti Zelati, M. Sorella, and D. Villringer. *Alpha-unstable flows and the fast dynamo problem*. *Communications on Pure and Applied Mathematics*, accepted for publication. arXiv:2504.00855.

Accepted M. Coti Zelati, M. Hairer, and D. Villringer. *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise*. *Indiana University Mathematics Journal*, accepted for publication. arXiv:2507.11422.

2025 D. Villringer. *Enhanced dissipation via the Malliavin calculus*. *Electronic Communications in Probability* **30** (2025), 1–11. doi:10.1214/25-ECP676.

Preprints

2026 M. Coti Zelati, M. Sorella, and D. Villringer. *Smooth autonomous fast dynamo action on the three-torus*. Preprint. arXiv:2609.04153.

2026 M. Coti Zelati, M. Sorella, and D. Villringer. *A fast dynamo on the three-torus*. Preprint. arXiv:2603.09861.

2025 M. Sorella and D. Villringer. *A limsup fast dynamo on* $\mathbb{T}^3$. Preprint. arXiv:2511.23024.

Prizes and awards

2025	**Doris Chen Merit Award**	Imperial College London
2022	**Imperial College President's PhD Scholarship**	Imperial College London
2022	**Senior Scholarship**	Trinity College, Cambridge
2022	**Examination Prize**	Trinity College, Cambridge
2021	**Trinity External Research Studentship**	Trinity College, Cambridge
2021	**Wynne Roberts Prize**	University College London
2021	**IMA Prize**	University College London
2021	**Mathematical and Physical Sciences Dean's List**	University College London
2020	**Kestelman Second Year Prize**	University College London
2019	**Undergraduate First Year Prize**	University College London

Seminars and conference talks

25 Sep 2026	**Analysis Seminar**	EPFL
10 Aug 2026	**Stable and unstable dynamics in PDEs and Hamiltonian systems**	Casa Matemática Oaxaca
27 Jan 2026	**2nd Münster–Imperial Day in PDEs**	University of Münster
2025	**Junior Dynamics Seminar**	Imperial College London
15 Sep 2025	**Long time dynamics in random and deterministic systems**	Bernoulli Center, EPFL
2025	**Blurring the lines between pure and applied through mixing**	Sabhal Mòr Ostaig
6 Nov 2024	**CDT Workshop**	Mathematics of Random Systems
2024	**Junior Analysis Seminar**	Imperial College London

Teaching experience

		EPFL
Autumn 2026	**Analyse I (allemand)**	Teaching Assistant
		Imperial College London
2023, 2025	**Analysis 2**	Teaching Assistant
2024	**Lebesgue Measure and Integration**	Senior Teaching Assistant
2023	**Lebesgue Measure and Integration**	Teaching Assistant
2022–2023	**Analysis 1**	Teaching Assistant

Languages

German: native English: bilingual Spanish: elementary